

Music-Gen v4 — Cycles 69-71

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Music-Gen v4 — Cycles 69-71

Abstract

Cycles 69-71 closed the non-Chicken-Grease drums arc at 4/4 SF2_CONFIRMED, completed the family-1 sibling-driver triad (bass at c1 + piano at Cycle 70 + other at Cycle 71) under a codified `sweep_driver_family_policy_*.md` policy landed one cycle before each driver, retired two profile-matrix cells via twin SKIP auto-closes, and correctly escalated the WIG piano stage-1 launch to operator authority after a legitimate one-cycle deferral surfaced a genuine architectural blocker on disk pressure that agent action cannot resolve. Cycle 69 landed the Peach Dream drums verdict SF2_CONFIRMED (ledger event ca668379-...) on the stage-2 leaderboard fed by the prior range's Cycle 68 detached launch, closing the non-CG drums arc at 4/4 and completing the c47-omnibus-enabled substantive execution phase's second-arc closure (after the non-CG bass arc closed at 4/4 in the prior range). The OP-2 Monitor pre-registration discipline was established here as first-tool-call-of-cycle practice. Cycle 70 attempted WIG piano stage-1 launch under the newly-authored `coarse_sweep_sf2_piano.py` sibling driver but was blocked by the 82% disk-pressure precondition; deferred once (legitimate first-deferral under the prior audit's carry-allowance) with concrete resume conditions, landed `coarse_sweep_sf2_piano.py` plus its test suite plus the codified `sweep_driver_family_policy_piano_c59.md` policy document, and the Cycle 70 auditor issued explicit guidance: "do NOT defer a second cycle without escalation." Cycle 70 also introduced an AST-only test-04 refinement on the piano test suite (identifier match rather than string match) that would propagate to the other-family tests one cycle later. Cycle 71 discharged five priorities: on P1, the worker probed the c60-brief prune target (c31_smoke/guitar_fine_legacy, 946 MB expected) and discovered the target absent on-disk — nearest present artifact c30_smoke = 4.0 KB tombstone — with actual disk pressure sitting on READ-ONLY v3/v3_spine anchors that FD-1 forbids unilateral agent deletion of, and did NOT fabricate an alternate prune to satisfy the brief's letter but disclosed the divergence honestly per invariant (d) and escalated with four named operator paths (OPT_A execute WIG piano stage-1 launch; OPT_B amend the 82% disk-pressure precondition; OPT_C widen the path-deletion policy; OPT_D revisit the precondition itself given ~7.5 GB freed on 252 GB with 5.8 GB avail is arithmetically infeasible without operator-scoped decisions), emitting `_plan/wig-piano-stage1-escalation-`

c61 id 69f293a9-... with authority=OPERATOR, blocked_on_operator=true, supersedes_path string → _plan/wig-piano-stage1-launch-deferred-c60 per the c14 lemma. This is exactly the halt-honest escalation shape the c60 auditor's mandate required, and it is forward-executable (concrete df readout, prune inventory, four named operator resolutions) rather than a wait-on-operator-memo violation. On P2, the other-family sibling driver coarse_sweep_sf2_other.py (SHA f6f81f4393e5ad00e04a054c5e56744426ce3ccb965a519f2fad88ed1c2b4bd4) landed policy-compliant with its 8/8 PASS test suite (SHA 54d5623a3f6c210e4332b4e4cc0ac65933916482dbf5f11ba741829378a80a26) under the codified sweep_driver_family_policy_other_c60.md (55be79b8...) OPT_B lane, with preset defaults bank0:programs=48,49,52,88,89,90,95,96 matching the policy's Recommended presets and parent + other policy SHAs both emitted in the manifest per the policy's \$Layered compliance clause, and the c60-introduced AST-only test-04 refinement propagated from piano tests to other tests. On P3, twin SKIP auto-closes retired two profile-matrix cells cleanly: vocals-family-skip-auto-close-c61 id 3bee87ed-... under FD-6 authority; guitar-family-1-skip-auto-close-c61 id 4cd4b57b-... under c15 SF2_RULED_OUT + c47 OPT1 extension. On P4, housekeeping close landed three events (_run/cycle_61_closed d41e6222-..., _archive/cycle-61-scratch 2c597e1f-..., _infra/adopt-cycle61-tests cdd67c72-...); all 8 read-only anchors byte-identical pre-vs-post. On P5, the §5 closing-summary contract rendered verbatim for the fourth consecutive cycle (c58 → c59 → c60 → c61) with 9 headers in exact order, _none this cycle_ placeholders where warranted, and no inter-section prose. Independent audit returned **VALIDATED** across all sub-topic assessment gates; the auditor recommended the next-cycle brief drop the inline closing-summary reminder since the pattern is now durable practice.

Introduction

The Music-Gen v4 closure campaign is driving through seven strictly-ordered milestones toward a clean close. The prior range advanced the non-CG drums arc from 0/4 to 3/4 SF2_CONFIRMED (WIG, Disco A, Rome) and launched the Peach Dream drums stage-2 detached with the verdict deferred to the next cycle per brief carry-allowance. The prior range also honestly deferred the WIG piano stage-1 launch with a two-point rationale (disk near c27 prune threshold; coarse_sweep_sf2.py docstring not yet validated for piano-family use) and enumerated two concrete resume options (OPT_A audit for instrument-agnostic parameterization; OPT_B author dedicated coarse_sweep_sf2_piano.py sibling per c10/c11/c13 pattern).

Cycles 69-71 are the range in which the non-CG drums arc closes, the family-1 sibling-driver triad completes under a newly-codified policy pattern, two profile-matrix cells retire cleanly under authority chains, and the piano launch discovers a genuine architectural blocker that the campaign correctly surfaces to operator authority rather than tries to defer around a second time. The range also stabilizes the closing-summary contract discipline at four consecutive compliant cycles, at which point the auditor judges the pattern durable enough to drop the inline reminder.

Approach

Cycle 69 (PD drums closure; OP-2 Monitor pre-registration discipline). Registered OP-2 Monitor task on the Peach Dream stage-2 log path as the first tool call of the cycle at cycle open — establishing the first-tool-call-of-worker-turn discipline the prior range's audit had recommended. On stage-2 DONE: observed, emitted the standard Peach Dream drums triple (drums.json with invariant (d) disclosure of non-standard operator_section_c25_checkpointed/rc9_6stem/ stem path per c19 stem_manifest.json SHA c4944ee80..., drums.replay_proof.json via c11 channel-aware replay.py with channel 10, drums_family_verdict.json update at song scope), and landed _lands/pd-drums-sf2-confirmed-c59 event id ca668379-.... Non-CG drums arc closes at 4/4 SF2_CONFIRMED — matches the non-CG bass 4/4 closure pattern from the c55 range.

Cycle 70 (family-1 driver triad opens on piano; legitimate first-deferral; policy codification pattern established). Landed `coarse_sweep_sf2_piano.py` as the family-1 sibling driver for piano-family sweeps (per the prior range’s `OPT_B` recommendation, since the `coarse_sweep_sf2.py` docstring was scoped to “bass” and audit for instrument-agnostic parameterization did not clear). Codified the sibling-driver architecture in `sweep_driver_family_policy_piano_c59.md` — a policy document that pins the additive lane semantics, preset defaults, layered-compliance-manifest expectations, and test-suite shape for family-1 sibling drivers. Attempted WIG piano stage-1 launch under the new driver but was blocked by the 82% disk-pressure precondition; deferred once (legitimate first-deferral under the prior audit’s carry-allowance) with concrete resume conditions naming a specific prune target (`c31_smoke/guitar_fine_legacy`, expected ~946 MB). Introduced an AST-only test-04 refinement on the piano test suite (identifier match rather than string match) — a refinement that would propagate to the other-family tests one cycle later. Cycle 70 auditor issued explicit forward guidance: “do NOT defer a second cycle without escalation.”

Cycle 71 (five-priority execution; P1 escalation is the defining artifact).

- **Priority 1 (WIG piano stage-1 resume-or-escalate).** Probed the `c60-brief` prune target `c31_smoke/guitar_fine_legacy` (expected 946 MB) and discovered the target ABSENT on-disk — the nearest present artifact `c30_smoke` was a 4.0 KB tombstone. Actual disk pressure sits on READ-ONLY `v3/v3_spine` anchors that FD-1 forbids unilateral agent deletion of. Worker did NOT fabricate an alternate prune target to satisfy the brief’s letter; instead disclosed the divergence per invariant (d) with concrete `df` readout and prune-inventory evidence, and escalated per the Cycle 70 auditor’s explicit mandate. Escalation event `_plan/wig-piano-stage1-escalation-c61 id 69f293a9-...` enumerated four named operator paths:
 - **OPT_A** — execute WIG piano stage-1 launch under existing disk margin (operator judgment on acceptable risk).
 - **OPT_B** — amend the 82% disk-pressure precondition (revisit the threshold in light of actual accessible-space arithmetic).
 - **OPT_C** — widen the path-deletion policy so agent can prune `v3/v3_spine` anchors under operator-scoped exemption.
 - **OPT_D** — revisit the 82% precondition itself, since freeing ~7.5 GB on 252 GB with 5.8 GB available is arithmetically infeasible without operator-scoped decisions.
 Event fields: `authority=OPERATOR`, `blocked_on_operator=true`, `supersedes_path` string → `_plan/wig-piano-stage1-launch-deferred-c60` per the `c14` lemma.
- **Priority 2 (other-family sibling driver + tests).** Landed `coarse_sweep_sf2_other.py` (SHA `f6f81f4393e5ad00e04a054c5e56744426ce3ccb965a519f2fad88ed1c2b4bd4`) with 8/8 PASS test suite (SHA `54d5623a3f6c210e4332b4e4cc0ac65933916482dbf5f11ba741829378a80a26`) policy-compliant per `sweep_driver_family_policy_other_c60.md` (`55be79b8...`) `OPT_B` lane. Preset defaults `bank0:programs=48,49,52,88,89,90,95,96` match the policy’s \$Recommended presets. Parent + other policy SHAs both emitted in the manifest per the policy’s \$Layered compliance clause. AST-only match refinement propagated from Cycle 70’s test-04 to the other-family test suite.
- **Priority 3 (twin SKIP auto-closes).** Two profile-matrix cells retired cleanly ledger-only, no driver invocation, no audio work:
 - `vocals-family-skip-auto-close-c61 id 3bee87ed-...` under FD-6 authority (operator ear is LANDS authority; vocals family has no non-operator-ear substitute).
 - `guitar-family-1-skip-auto-close-c61 id 4cd4b57b-...` under `c15 SF2_RULED_OUT` + `c47 OPT1` extension (family-1 guitar was ruled out at `c15` for CG under the pre-`c47` policy; `c47 OPT1` lift does not extend to a family that was already ruled out on family-1 semantics).

- **Priority 4 (housekeeping close).** Three events landed: `_run/cycle_61_closed id d41e6222-...`; `_archive/cycle-61-scratch id 2c597e1f-...`; `_infra/adopt-cycle61-tests id cdd67c72-...`. All 8 read-only anchors verified byte-identical pre-vs-post.
- **Priority 5 (§5 closing-summary contract).** 9 headers rendered verbatim in exact order at exact heading levels; `_none` this cycle placeholders where warranted; no inter-section prose. **Fourth consecutive compliant cycle** (c58 → c59 → c60 → c61).

Discipline guards asserted across the range. OP-2 Monitor pre-registration at first tool call of each cycle (established c59; sustained through the range close; Cycle 71’s registration timestamp 2026-09-05T17:59:03Z per §3.1 mandate). Zero `_manager/*` escalation memos re-opened; all six c47 omnibus-closed operator-authority memos remain CLOSED. The Cycle 71 P1 escalation is a `_plan/` event (agent-initiated forward-executable escalation), not a `_manager/` event (which would be a new operator-authority surface). No preservation-spin subleaves (fourteen consecutive clean cycles since c48). No wait-on-operator memo (banned per operator directive 2026-09-03 point 2); the Cycle 71 P1 escalation is forward-executable, not a wait-on-operator memo, because it enumerates four concrete operator paths with the specific state each would produce. `c14` string-supersedes_path lemma honored on each new emission. All AST-scannable invariants pass: no PRNG, no sidecar_nonfactor, no VST3 state APIs, no `--verify-det` bypass, `/usr/bin/python3` interpreter guard. `SF2_CONFIRMED` emitted under c47 OPT1-extended acceptance on Peach Dream drums (c59) with no staging through provisional. Ledger delta 7 events at Cycle 71 (4 substantive + 3 housekeeping) meets the §3.7 ≥ 3 target.

Findings

Non-CG drums arc closed at 4/4 SF2_CONFIRMED

Cycle 69 emitted the Peach Dream drums verdict `SF2_CONFIRMED` (ledger event `ca668379-...`) on the stage-2 leaderboard fed by the prior range’s detached launch. The pinned triple (`drums.json` + `drums.replay_proof.json` + `drums_family_verdict.json`) landed at `data/v4/profiles/88d247468cb6d49f/` with invariant (d) disclosure of the non-standard `operator_section_c25_checkpointed/rc9_6stem/` stem path per `c19` `stem_manifest.json` SHA `c4944ee80...`, and with the `c11` channel-aware `replay.py` (channel 10 for drums) generating the replay proof. Non-CG drums arc closes at 4/4 — matches the non-CG bass 4/4 closure pattern from c55. Four cells landed across four cycles (WIG c66; Disco A c66; Rome c68; Peach Dream c69 by external numbering; internal `c57 / c57 / c58 / c59`).

Family-1 sibling-driver triad complete

The family-1 sibling-driver triad now spans three complete drivers under a codified policy pattern:

- Bass: `coarse_sweep_sf2.py` at `c1` anchor.
- Piano: `coarse_sweep_sf2_piano.py` landed Cycle 70 under `sweep_driver_family_policy_piano_c59.md`.
- Other: `coarse_sweep_sf2_other.py` landed Cycle 71 (SHA `f6f81f4393e5ad00e04a054c5e56744426ce3ccb965a519f21`) under `sweep_driver_family_policy_other_c60.md` (`55be79b8...`); 8/8 PASS test suite (SHA `54d5623a...`); preset defaults `bank0:programs=48,49,52,88,89,90,95,96` matching the policy’s \$Recommended presets; parent + other policy SHAs both emitted in the manifest per the policy’s \$Layered compliance clause; AST-only test-04 refinement propagated from Cycle 70’s piano tests.

This is the third consecutive cycle in which a policy document landed at cycle N-1 governs an artifact at cycle N without mid-cycle architectural discussion — the “codify-before-execute”

pattern established across c59 / c60 / c61 (internal numbering). If a fourth family-driver landing repeats the same shape, the pattern earns family-policy invariant codification.

Cycle 71 P1 escalation is the range's defining artifact

The escalation is done correctly on every dimension the Cycle 70 auditor's mandate required:

- **Not deferred a second time.** The Cycle 70 auditor had explicitly banned a second deferral: "do NOT defer a second cycle without escalation." Cycle 71 escalated.
- **Not fabricated around the brief-vs-disk divergence.** When the c60-brief prune target c31_smoke/guitar_fine_legacy (expected 946 MB) was found absent on-disk (nearest present artifact c30_smoke = 4.0 KB tombstone), the worker did not substitute an alternate prune target to satisfy the brief's letter. Instead the worker disclosed the divergence per invariant (d) and named the FD-1 constraint (actual disk pressure sits on READ-ONLY v3/v3_spine anchors that FD-1 forbids unilateral agent deletion of).
- **Forward-executable, not a wait-on-operator memo.** The escalation enumerates four concrete operator paths (OPT_A execute launch; OPT_B amend precondition; OPT_C widen deletion policy; OPT_D revisit precondition given arithmetic infeasibility), each with specific downstream state. It does not ask the operator to think about the problem; it asks the operator to choose one of four resolutions.
- **c14 lemma honored.** `supersedes_path` is a string pointing at the Cycle 70 deferral event (`_plan/wig-piano-stage1-launch-deferred-c60`), not a list.
- **Fields set correctly.** `authority=OPERATOR`, `blocked_on_operator=true` — the escalation is on the correct authority surface.

Event id `_plan/wig-piano-stage1-escalation-c61` = 69f293a9-....

Twin SKIP auto-closes retire two profile-matrix cells cleanly

Both SKIP auto-closes are ledger-only with no phantom work:

- `vocals-family-skip-auto-close-c61` id 3bee87ed-... under FD-6 authority. Vocals family has no non-operator-ear substitute; the SKIP is the correct auto-closure per the operator-ear-only-LANDS invariant.
- `guitar-family-1-skip-auto-close-c61` id 4cd4b57b-... under c15 SF2_RULED_OUT + c47 OPT1 extension. Family-1 guitar was ruled out at c15 for CG under the pre-c47 policy; the c47 OPT1 lift does not extend to a family that was already ruled out on family-1 semantics — the SKIP correctly recognizes the c15 ruling as terminal for the family-1 lane.

Closing-summary contract at fourth consecutive compliant cycle

Rendered verbatim per §5 template with 9 headers in exact order at exact heading levels, `_none` this cycle placeholders where warranted, no inter-section prose. Fourth consecutive compliant cycle across c58 → c59 → c60 → c61 (internal numbering). The inline-enforcement bet from the Cycle 67 audit (external numbering; c57 internal) is now durable practice. The Cycle 71 auditor recommends the next-cycle brief drop the explicit inline reminder — the pattern is stable enough to trust without in-brief re-emphasis; if a future auditor observes drift, the reminder can be restored.

Read-only anchors held; discipline invariants met

All 8 read-only anchors verified byte-identical pre-vs-post at range close. No drift detected; no invariant-(d) SHA disclosures needed for the anchors themselves (the only invariant-(d)

disclosure this range concerns the absent prune target on the P1 escalation). Canonical 7-key env_pin_sha256=2ac444c3...a922ca stands from prior range; FD-16(a) re-issue not triggered. OP-1 SerialLock engaged and released cleanly on the c69 Peach Dream drums verdict emission. Six c47 omnibus-closed operator-authority memos remain CLOSED throughout.

Audit outcome

VALIDATED across all sub-topic assessment gates. Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. The P1 escalation is the range's defining artifact and is done correctly on every discipline dimension. P2 completes the family-1 driver triad under a policy landed one cycle prior — a stable codify-before-execute cadence. P3 twin SKIPS retire two profile-matrix cells with correct provenance chains. P4 housekeeping close clean with all read-only anchors byte-identical. P5 closing-summary contract at fourth consecutive compliant cycle.

Discussion

Three things about this range are worth naming.

First, the Cycle 71 P1 escalation is a discipline-mechanism example worth preserving. The prior audit's mandate ("legitimate deferral once, escalation on second cycle if the underlying blocker persists") gave the worker a specific decision to execute, and the worker executed it in a way that satisfied every discipline invariant the mandate implied. When the worker probed the brief-specified prune target and found it absent from disk, three shortcut paths were available: fabricate an alternate prune target to satisfy the brief's letter; defer a second cycle in violation of the mandate; or open a `_manager/*` wait-on-operator memo (banned per operator directive). Instead the worker chose the fourth path — honest divergence disclosure per invariant (d), FD-1 recognition that agent action cannot resolve the underlying constraint, and a forward-executable escalation event with four named operator paths and specific downstream state per path. The escalation is not "operator please decide" (which would violate the wait-on-operator memo ban); it is "operator, here are four resolutions each of which produces a specific next-cycle state — pick one." This is the correct shape of halt-honest escalation under the c47-era discipline invariants.

Second, the "codify-before-execute" pattern has now run three consecutive cycles (c59 / c60 / c61 internal) and is stable enough to be named. The pattern: a policy document lands at cycle N-1 codifying the additive lane semantics, preset defaults, layered-compliance-manifest expectations, and test-suite shape for a class of sibling-driver landings; the artifact lands at cycle N and its manifest cites the parent-plus-family policy SHAs per the policy's `$Layered` compliance clause. c60's piano policy governed c61's piano driver landing; c61's other policy governed c62's other driver landing. If a fourth family-driver landing in a subsequent cycle repeats the same shape, the pattern earns family-policy invariant codification in `docs/agent_picks_selection_invariants.md`. The pattern is worth preserving because it separates the architectural discussion from the artifact execution — the discussion happens one cycle in advance and produces a concrete policy document that the following cycle's worker can execute against without re-deriving the semantics.

Third, the closing-summary contract's transition from inline-enforced to durable-practice is a discipline-mechanism completion worth naming. The pattern started three ranges ago as an M-1 discipline-drift observation (worker closing summaries not rendering the mandated template); the audit made an inline-enforcement bet (codify the template verbatim in the next brief; break the drift without a `_manager/*` escalation memo); the discipline stabilized in one cycle; three more cycles held the pattern; the current range's audit judges the pattern durable enough to drop the inline reminder from the next brief. This is the correct terminal

state for an inline-enforcement bet — the reminder was scaffolding for a discipline that has now internalized, and the scaffolding should be removed to keep the brief lean, with the audit checklist retaining the contract as a first-line check if a future drift surfaces. If drift returns, the reminder returns. The mechanism completes gracefully.

Open questions

- **Operator adjudication of Cycle 71 P1 escalation.** `_plan/wig-piano-stage1-escalation-c61 id 69f293a9-...` names four concrete operator paths (OPT_A execute launch; OPT_B amend precondition; OPT_C widen deletion policy; OPT_D revisit precondition). Next-cycle P0 (BLOCKING) is to check `live_guidance` for operator adjudication. If operator adjudicates, execute the chosen path as next-cycle P1 first-act. If operator does NOT adjudicate, next-cycle P1 is NOT re-escalation (that would violate the c60 auditor’s “do NOT defer without escalation” rule at the escalation level — you cannot escalate an escalation); instead either advance P2 (other-family sweep launch, gated on disk clearance) if disk clears organically, or enter genuine `blocked_on_operator` state for WIG piano only while progressing other work. Do NOT fabricate a prune target that does not exist. Do NOT defer P1 to the cycle after next.
- **OP-2 Monitor pre-registration discipline.** Sustained across the range; recommend continued as first-tool-call-of-worker-turn practice with ToolSearch load mandate if not warm.
- **Other-family sweep launch prep.** Cycle 71 landed the driver + tests; next cycle can launch the other-family stage-1 sweep under Monitor if disk pressure resolves. Otherwise honestly defer with concrete resume command.
- **Operator directive #5(c) queue enumeration.** Post-vocals + guitar-family-1 auto-closes, the remaining audible-stems queue needs explicit brief enumeration (drums + bass + guitar-family-2 across 4 non-CG focus songs, minus the closed arcs and minus the auto-closed cells). Enumerate in the next-cycle brief so the worker does not re-derive.
- **§5 closing-summary contract inline reminder.** Auditor recommends dropping the inline reminder from the next-cycle brief. Four consecutive compliant cycles (c58 → c59 → c60 → c61 internal) is sufficient basis to trust the pattern. Keep the contract itself in the audit checklist; remove the explicit inline reminder from the brief. If a future auditor observes drift, restore the reminder.
- **AST-only test-04 refinement promotion.** Second cycle carrying this refinement (Cycle 70 piano; Cycle 71 other). If it recurs for a third family, promote to a family-policy invariant.
- **Ledger delta target.** If the next-cycle P0 is fully operator-blocked and P2 is disk-blocked, achieving the ≥ 3 substantive-event target becomes challenging. Legitimate low-ledger-count cycles are honest under FD-1; the next-cycle brief should explicitly permit this rather than manufacture events.
- **Non-CG SF2_CONFIRMED terminal status.** Bass 4/4 CLOSED (c55 range); Drums 4/4 CLOSED (c59 PD event id `ca668379-...`). Do not re-open.
- **Downstream sequence per operator directive #5.** With 4/4 non-CG bass done, 4/4 non-CG drums done, family-1 driver triad complete, vocals + guitar-family-1 auto-closed, remaining work under (c) is guitar-family-2 per applicable songs and piano / other per applicable songs; then (d) re-render + deliver A/B per song using pinned profiles; (e) fresh generator batch; (f) amended completion report; (g) clean re-close.

Appendix: Provenance

Directive. Execute the Music-Gen v4 closure campaign; pursue milestones in strict order starting with M-V4-CERT-1 and M-V4-PROFILES-1; drive to a clean close.

Cycle range. cycles 69-71.

Working directory. /home/user/long-exposure-runs/music-gen.

Session references.

- Cycle 69 researcher aa34507a-a7ae-461a-9cc4-3f5c1d04b3b3; worker 16c30409-43fe-4704-bf04-3259528fce54; auditor 27bddd47-4862-4aed-95e6-1c8878a41a37.
- Cycle 70 researcher 5a6fcbac-8814-4de4-8073-9ba1bdcf5920; worker b380cb6c-59cc-4376-9d02-429ac66fa924; auditor a4de1345-aed0-4d6f-a079-4d9f525a42d6.
- Cycle 71 researcher c981315a-622a-4e1e-a678-7ccc620158d5; worker e5679fcf-9a8d-48c7-b67f-0f91efdd55bf; auditor 65cb72ff-2538-437a-b388-bcb16db676dd.

Audit verdict. **VALIDATED** across all sub-topic assessment gates. Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR.

Terminal deliverables landed this range.

- **Peach Dream drums SF2_CONFIRMED (Cycle 69).** Ledger event _lands/pd-drums-sf2-confirmed-c59 id ca668379-.... Pinned triple at data/v4/profiles/88d247468cb6d49f/ with invariant-(d) stem-path disclosure per c19 stem_manifest.json c4944ee80-.... Non-CG drums arc closes at 4/4.
- **coarse_sweep_sf2_piano.py family-1 sibling driver + sweep_driver_family_policy_piano_c59.md policy (Cycle 70).** OPT_B lane per prior range's recommendation. AST-only test-04 refinement introduced on piano test suite.
- **WIG piano stage-1 legitimate first-deferral (Cycle 70).** _plan/wig-piano-stage1-launch-deferred-c60. Concrete resume conditions named specific prune target c31_smoke/guitar_fine_legacy (~946 MB expected).
- **coarse_sweep_sf2_other.py family-1 sibling driver + sweep_driver_family_policy_other_c60.md policy (Cycle 71).** Driver SHA f6f81f4393e5ad00e04a054c5e56744426ce3ccb965a519f2fad88ed1c2b4bd4; test suite SHA 54d5623a3f6c210e4332b4e4cc0ac65933916482dbf5f11ba741829378a80a26 8/8 PASS; policy SHA 55be79b8-...; preset defaults bank0:programs=48,49,52,88,89,90,95,96; parent + other policy SHAs in manifest per \$Layered compliance. AST-only test-04 refinement propagated from piano tests.
- **WIG piano stage-1 escalation (Cycle 71).** _plan/wig-piano-stage1-escalation-c61 id 69f293a9-...; authority=OPERATOR, blocked_on_operator=true, supersedes_path string → _plan/wig-piano-stage1-launch-deferred-c60 per c14 lemma; four named operator paths (OPT_A execute launch; OPT_B amend 82% precondition; OPT_C widen path-deletion policy; OPT_D revisit precondition given ~7.5 GB free-target on 252 GB with 5.8 GB available). Prune-target absence disclosed per invariant (d) with df readout + prune inventory + FD-1 recognition that actual disk pressure sits on READ-ONLY v3/v3_spine anchors.
- **Twin SKIP auto-closes (Cycle 71).** vocals-family-skip-auto-close-c61 id 3bee87ed-... (FD-6 authority); guitar-family-1-skip-auto-close-c61 id 4cd4b57b-... (c15 SF2_RULED_OUT + c47 OPT1 extension).
- **Housekeeping tail (Cycle 71).** _run/cycle_61_closed id d41e6222-...; _archive/cycle-61-scratch id 2c597e1f-...; _infra/adopt-cycle61-tests id cdd67c72-.... 7 events total (4 substantive + 3 housekeeping).

Six operator escalations remain formally closed on the substantive side. No _manager/* events opened this range. data/v4/_manager/ state untouched. The Cycle 71 P1 escalation is a _plan/ event (agent-initiated forward-executable escalation), not a _manager/ event.

Read-only anchors preserved byte-identical pre-vs-post (8/8 verified at range close).

Discipline surface unchanged this range across the standard set (scripts/sound_match/objective.py 8087ce80-...; scripts/sound_match/_sweep_hygiene_c27.py 771ff42b-...; scripts/sound_match/_serial_lock_

opl.py b8e1b7dda5dled19...; data/v4/deliveries/31a164f845f8e27e/cg_ab_mix.wav 6e13e007...; the four policy-and-invariants documents including docs/agent_picks_selection_invariants.md 29a1610b... and docs/emitter_exemption_policy.md fd2c33a7...).

Environment pin. Canonical 7-key env_pin_sha256=2ac444c36298d6ada0579aba1a9160a5881703a4e628f5cccd828b842 stands from prior range; FD-16(a) re-issue not triggered.

Discipline guards asserted. OP-2 Monitor pre-registration at first tool call of each cycle (Cycle 71 timestamp 2026-09-05T17:59:03Z per §3.1 mandate). Zero _manager/* escalation memos re-opened; all six c47 omnibus-closed operator-authority memos remain CLOSED. No preservation-spin sub-leaves (14 consecutive clean cycles since c48). No wait-on-operator memo (banned per operator directive 2026-09-03 point 2); the Cycle 71 P1 escalation is forward-executable per its four named operator paths. c14 string-supersedes_path lemma honored throughout. All AST-scannable invariants pass: no PRNG, no sidecar_nonfactor, no VST3 state APIs, no --verify-det bypass, /usr/bin/python3 interpreter guard. SF2_CONFIRMED on Peach Dream drums emitted under c47 OPT1-extended acceptance with no staging through provisional. Fourth consecutive §5 closing-summary contract compliance across c58 → c59 → c60 → c61 internal numbering.

Milestone status at range close.

- M-V4-CERT-1 — validated (E2E_DETERMINISM_HOLDS on the v3 spine).
- M-V4-PROFILES-1 CG (5/5 instruments) — validated (unchanged).
- M-V4-PROFILES-1 non-CG bass — 4/4 SF2_CONFIRMED (CLOSED prior range; unchanged).
- **M-V4-PROFILES-1 non-CG drums — 4/4 SF2_CONFIRMED (CLOSED this range at Cycle 69).**
- M-V4-PROFILES-1 non-CG vocals — SKIP auto-closed under FD-6 authority (Cycle 71).
- M-V4-PROFILES-1 non-CG guitar family-1 — SKIP auto-closed under c15 SF2_RULED_OUT + c47 OPT1 extension (Cycle 71).
- M-V4-PROFILES-1 non-CG guitar family-2 — queued per operator directive #5(c).
- M-V4-PROFILES-1 non-CG piano — WIG piano stage-1 launch escalated to operator authority (Cycle 71 P1); driver + policy landed (Cycle 70).
- M-V4-PROFILES-1 non-CG other — driver + policy landed (Cycle 71); stage-1 launch queued gated on disk pressure clearance.
- M-V4-SHOWCASE-1 CG — LANDS_pending_operator (cg_ab_mix.wav byte-identical since c17).
- M-V4-SHOWCASE-1 non-CG — unblocked at policy level; A/B deliveries queued per operator directive #5(d) using the pinned non-CG bass + drums profiles now available.
- M-V4-RULES-1 — scaffold landed c20; substantive implementation queued.
- M-V4-EAR-1 — not yet opened.
- M-V4-GEN-1 — conditional on M-V4-RULES + M-V4-EAR; queued for fresh stall-budget-reset batch per operator directive #5(e).
- M-V4-CLOSE-1 — c24 amendment landed; completion report v3 queued per operator directive #5(f).

Substantive execution cadence. Seven consecutive successful substantive cycles under the c47 omnibus post-preservation-cadence era (c55 → c61 internal numbering). Positive overshoot pattern from c55 sustains.

Next-cycle first tasks (per auditor forward guidance). (a) P0 BLOCKING check live_guidance for operator adjudication of _plan/wig-piano-stage1-escalation-c61 id 69f293a9-.... If operator adjudicates, execute chosen path as P1 first-act. If not, P1 is NOT re-escalation; either advance P2 (other-family sweep launch, gated on disk clearance) if disk clears organically, or enter genuine blocked_on_operator state for WIG piano only while progressing other work.

(b) OP-2 Monitor registration as first tool action of worker turn per §3.1 mandate, loading via ToolSearch if not warm. (c) Other-family sweep launch prep if disk allows; otherwise honestly defer with concrete resume command. (d) Enumerate operator directive #5(c) remaining audible-stems queue in the brief so worker does not re-derive. (e) Drop the inline closing-summary contract reminder from §5 of the next brief; keep the contract in the audit checklist. (f) Legitimate low-ledger-count cycles are honest under FD-1 if P0 operator-blocked and P2 disk-blocked; brief should explicitly permit this rather than manufacture events. Operator ear remains LANDS authority post-hoc per FD-6.